rnaseq-nf pipeline

```
user125444@MSI: ~/rnaseq_project
(base) user125444@MSI:~/rnaseq_project$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

(base) user125444@MSI:~/rnaseq_project$ nextflow pull nextflow-io/rnaseq-nf
Checking nextflow-io/rnaseq-nf ...
 downloaded from https://github.com/nextflow-io/rnaseq-nf.git - revision: ca20a6dfd2
[master]
(base) user125444@MSI:~/rnaseq_project$
```

Run the pipeline on the test data in Docker

```
(base) user125444@MSI:~/rnaseq_project$ nextflow run nextflow-io/rnaseq-nf -profile docker

N E X T F L O W  ~  version 25.10.4

Launching `https://github.com/nextflow-io/rnaseq-nf` [boring_kirch] DSL2 - revision: ca20a6dfd2 [ma
ster]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/user125444/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49
020000.Ggal71.500bpflank.fa
reads        : /home/user125444/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : results

executor >  local (4)
[29/eeeec6] RNA…NDEX (ggal_1_48850000_49020000) | 1 of 1 ✔
[d2/5e3229] RNASEQ:FASTQC (ggal_gut)            | 1 of 1 ✔
[52/34220c] RNASEQ:QUANT (ggal_gut)             | 1 of 1 ✔
[65/e7ffec] MULTIQC                             | 1 of 1 ✔

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 09-Apr-2026 20:56:24
Duration    : 3m 9s
CPU hours   : 0.1
Succeeded   : 4
```

multiqc_report.html

```
(base) user125444@MSI:~/rnaseq_project$ ls results
fastqc_ggal_gut_logs  multiqc_report.html
```

```
(base) user125444@MSI:~/rnaseq_project$ ls work
29  52  65  d2
```

Run the pipeline on the test data in Conda

```
(base) user125444@MSI:~/rnaseq_project$ conda activate rnaseq_env
(rnaseq_env) user125444@MSI:~/rnaseq_project$ nextflow run nextflow-io/rnaseq-nf -profile conda

 N E X T F L O W  ~  version 25.10.4

Launching `https://github.com/nextflow-io/rnaseq-nf` [distracted_varahamihira] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/user125444/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_1_48850000_49020000.Ggal71.500bpflank.fa
reads        : /home/user125444/.nextflow/assets/nextflow-io/rnaseq-nf/data/ggal/ggal_gut_{1,2}.fq
outdir       : results

executor >  local (4)
[fb/5659b1] RNASEQ:INDEX (ggal_1_48850000_49020000) [100%] 1 of 1 ✔
[c2/ec7060] RNASEQ:FASTQC (ggal_gut)                [100%] 1 of 1 ✔
[1e/a5a5c2] RNASEQ:QUANT (ggal_gut)                 [100%] 1 of 1 ✔
[ef/7750d4] MULTIQC                                 [100%] 1 of 1 ✔

Done! Open the following report in your browser --> results/multiqc_report.html

Completed at: 09-Apr-2026 21:14:34
Duration    : 1m 59s
CPU hours   : (a few seconds)
Succeeded   : 4
```

SRR292241

Dear Vitaly,

I have spent several hours trying to run the command provided on the Nextflow page in different variations, but they all failed. Could you please clarify what I might be doing wrong?

The command I used was:

nextflow run nextflow-io/rnaseq-nf --reads "/home/user125444/rnaseq_project/data/SRR292241_{1,2}.fastq.gz" --transcriptome "/home/user125444/rnaseq_project/data/gencode.v49.transcripts.fa" --outdir "/home/user125444/rnaseq_project/data/results_real" -profile docker

Only this variant started to run:

nextflow run nextflow-io/rnaseq-nf --reads "/home/user125444/rnaseq_project/data/*_*.fastq.gz" --transcriptome "/home/user125444/rnaseq_project/data/gencode.v49.transcripts.fa" --outdir "/home/user125444/rnaseq_project/data/results_real" -profile docker

```
(rnaseq_env) user125444@MSI:~/rnaseq_project/data$ nextflow run nextflow-io/rnaseq-nf --reads "/home/user125444/rnaseq_project/d
ta/*_*.fastq.gz" --transcriptome "/home/user125444/rnaseq_project/data/gencode.v49.transcripts.fa" --outdir "/home/user125444/rn
seq_project/data/results_real" -profile docker

N E X T F L O W  ~  version 25.10.4

Launching `https://github.com/nextflow-io/rnaseq-nf` [marvelous_tesla] DSL2 - revision: ca20a6dfd2 [master]

R N A S E Q - N F   P I P E L I N E
===================================
transcriptome: /home/user125444/rnaseq_project/data/gencode.v49.transcripts.fa
reads        : /home/user125444/rnaseq_project/data/*_*.fastq.gz
outdir       : /home/user125444/rnaseq_project/data/results_real

executor >  local (1)
[22/ef6d1d] RNASEQ:INDEX (gencode) [100%] 1 of 1
[-        ] RNASEQ:FASTQC          -
[-        ] RNASEQ:QUANT           -
[-        ] MULTIQC                -

Done! Open the following report in your browser --> /home/user125444/rnaseq_project/data/results_real/multiqc_report.html

Completed at: 09-Apr-2026 23:16:05
Duration    : 15m 5s
CPU hours   : 0.3
Succeeded   : 1
```

No report was created, and as far as I can see, no analysis has been done.

Since the paired-end files from the NCBI site are usually downloaded as a single file, could you please advise the correct procedure to split this file so that it can be used in Nextflow without errors?

I used the command seqkit split2 SRR292241.fastq.gz

Thank you.